
QEG-RAG Engine
Graph Retrieval & Controller
Backbone LLM
Neural Reasoning Engine
Tier 1: Initial Evidence Reasoning & Structural Clarity Audit
1. Primary Prompt: Evidence Multigraph + Guided Path Scaffolding
Full unpruned graph with linear traversal scaffold
Running Example: Toronto Eagles (1989) vs Toronto Blues (1989)
Question: Which university is affiliated with club formed in 1989?
Fast Exit
(≈ 80%)
1 LLM Call →
Verified Answer
2. Diagnostic Handoff: Candidate Collision Alert
Decision: Multiple clubs match 1989; missing Blues bridge doc
LLM stops without guessing → emits structured diagnostic packet
Automated Bridge Search (Zero LLM Calls)
1. Intercept Handoff:
Reads tied rows (Eagles, Blues)
2. Targeted Search (0 Calls):
Fetches Doc:Toronto_Blues
3. Assemble Capsule:
Prunes all irrelevant rows
Tier 2: Targeted Collision Resolution & Provenance Verification
3. Targeted Prompt: Minimal State Capsule
Focused context containing strictly tied candidates & retrieved bridge doc
Discards distracting noise → eliminates context dilution
Verified Answer: "York"
≤ 2 LLM Calls • Full Provenance